

PCD_Assignment01

Analysis Report | Two photographic inputs

Objective and input photographs

The objective is to compare Max, Average and Median downsampling with nearest-neighbor (NN), bilinear and bicubic upsampling. The assignment term “Medium” is interpreted as Median. Two student-supplied photographs are used. malvin.jpeg is a portrait photograph containing fine hair and fabric texture, smooth regions and geometric background edges. pemandangan.jpg is a landscape containing thin branches, vegetation, mountain ridges and smoother sky and cloud regions. Their contrasting spatial detail supports a practical comparison of information loss.

Preprocessing and implementation

The portrait is 1200 x 1600 pixels (width x height). The landscape is 474 x 842 pixels; two columns on the right and two rows at the bottom are removed, giving 472 x 840. This makes both dimensions divisible by four. No initial resize or stretching is applied. Images are oriented using EXIF metadata and converted to RGB. Downsampling replaces each non-overlapping 2 x 2 or 4 x 4 block with its maximum, arithmetic mean or median, independently for each channel. For even sample counts the median averages the two central sorted values. Results are rounded to uint8. Upsampling restores the processed original dimensions using the selected Pillow filter. Output PNG files avoid an additional JPEG compression step.

Experiment and evaluation

The notebook executes 36 pipelines: two photographs, two reduction factors, three pooling methods and three interpolation methods. The portrait reduces to 600 x 800 or 300 x 400; the landscape reduces to 236 x 420 or 118 x 210. Each reconstruction is compared with its processed original using MSE and $PSNR = 10 \log_{10}(255^2 / MSE)$. These scores evaluate the complete down-up pipeline. Higher PSNR means closer pixel agreement, not necessarily better perceived sharpness. All notebook code cells were executed locally in sequence, and their figures and outputs were saved. The supplied version used Python 3.12.14, NumPy 2.3.5 and Pillow 12.3.0. A signed-in Google Colab run is still required before submission.

Input / factor	Highest-scoring pipeline	MSE	PSNR
malvin / 2x	Average + Bicubic	21.62	34.78 dB
malvin / 4x	Average + Bicubic	73.51	29.47 dB
pemandangan / 2x	Average + Bicubic	338.40	22.84 dB
pemandangan / 4x	Average + Bicubic	779.06	19.22 dB

Table 1. Best measured pipeline for each input and scale. Full results: results/metrics.csv.

Effects of downsampling

Max pooling emphasizes brighter values in mixed blocks. The landscape crop makes the effect clear: bright sky expands into thin dark branches, and some narrow structures disappear. Max therefore does not preserve every kind of edge; the effect depends on contrast polarity. Average pooling blends each block and retains its mean intensity approximately, apart from rounding. It suppresses high-frequency detail, so fine strands and small twigs become indistinct. Median pooling can preserve a dominant dark or bright region, but minority features can be removed. In the landscape crop, it produces simplified, thicker-looking regions with missing fine branches. This experiment adds no artificial noise and has no clean noise-free reference, so it cannot establish a denoising ranking.

Effects of upsampling

NN visibly repeats the reduced samples as square blocks. Bilinear makes transitions smoother but softens narrow lines. Bicubic retains stronger transitions in the inspected crops than bilinear, although the missing original detail is still absent. Cubic interpolation can produce ringing near sharp boundaries; the experiment does not separately quantify ringing. At fourfold reduction, the portrait reconstruction no longer separates many fine strands visible in the original crop. In the landscape, broad branch shapes remain but fine twigs merge or vanish. Interpolation changes the appearance of the remaining information; it does not recover the discarded samples.

Numerical findings

Average + Bicubic achieves the highest original-reference PSNR among the nine tested pipelines for both photographs at both factors. For the portrait, PSNR falls from 34.78 dB at 2x to 29.47 dB at 4x. For the landscape it falls from 22.84 dB to 19.22 dB. All tested pipelines lose PSNR when moving from 2x to 4x on these images. Median + Bicubic is close on the portrait at 4x (29.40 dB), whereas Max + Bicubic is much lower (22.39 dB). Bicubic is not universally the highest-scoring interpolator for every pooling method: with Max at 4x, Bilinear slightly exceeds Bicubic on both photographs. These results support choosing a complete pipeline rather than assuming one interpolation filter always wins.

Conclusion and limitations

For these two inputs, Average downsampling followed by Bicubic upsampling gives the best pixel fidelity among the tested combinations. NN is useful for inspecting discrete pixel structure, while Bilinear provides smooth transitions. Stronger reduction sacrifices more texture and narrow edges. The landscape has lower scores in this experiment, consistent with its many fine high-contrast structures, but it also starts at a different resolution; this is not a controlled test of scene difficulty alone. JPEG artifacts in the supplied sources, RGB channel-wise processing, rounding, crop alignment and the lack of human quality ratings limit generalization. Block averaging provides box-filter smoothing, not ideal anti-aliasing; Max and Median are nonlinear and offer no general anti-aliasing guarantee. The conclusion applies to these images and parameters rather than all photographs.

Visual evidence: landscape detail

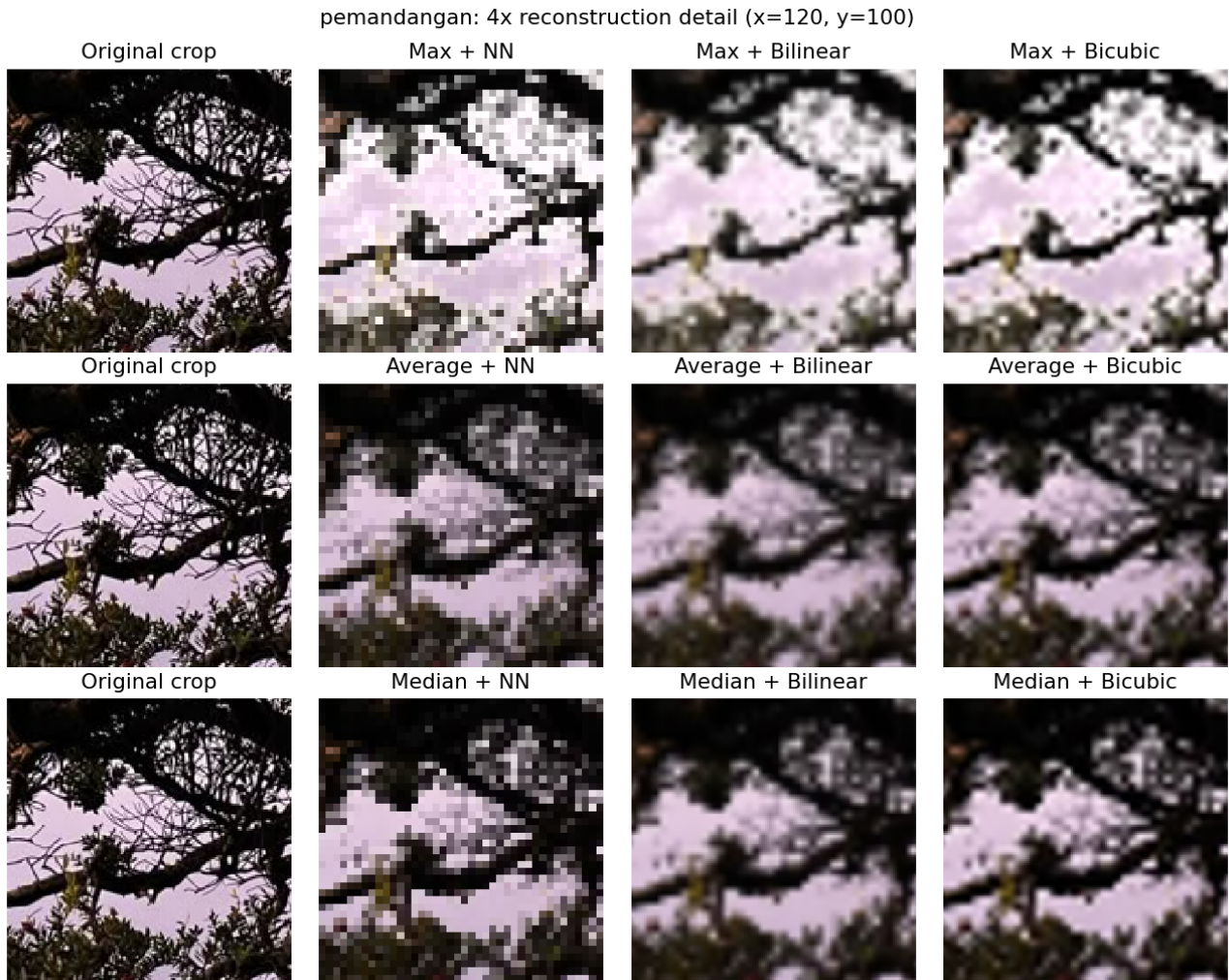


Figure 1. A 160 x 160 crop beginning at x=120, y=100, after 4x reduction and reconstruction. Each row repeats the original for direct comparison. Max expands bright areas, NN exposes blocks, and smooth interpolation cannot restore fine twigs. The display uses nearest-neighbor rendering to avoid adding smooth interpolation. A corresponding portrait detail figure is included in results/figures/malvin_detail.png.

Sources and reproducibility

Input files: malvin.jpeg and pemandangan.jpg, supplied by the student. Authorship or licensing of the photographs has not been independently established. All output images, measurements and comparisons were generated by the accompanying notebook.

Implementation reference: Pillow documentation, Concepts - Filters.

<https://pillow.readthedocs.io/en/stable/handbook/concepts.html#filters> (accessed September 12, 2026).